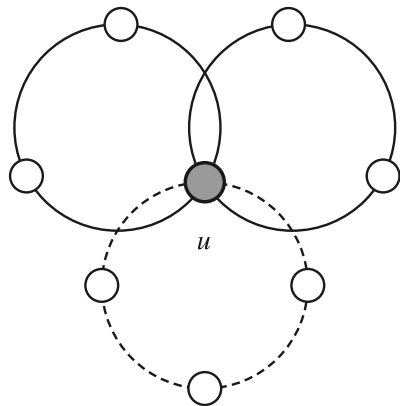


a

multiplex hypergraph

layer 1

 $m_1 = 3$ 

$$k^{(1)}(u) = 2$$

$$k^{(2)}(u) = 1$$

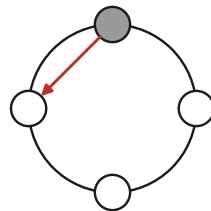
one state per node,
shared by all layers

layer 2

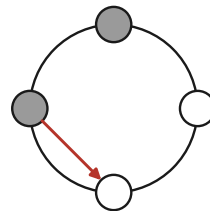
 $m_2 = 4$ **b**

intra-group cascade

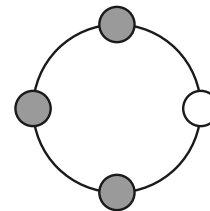
$$C(m, \lambda, \theta) > (m - 1) T$$



one seed



infects a member



which infects another

seed infectious

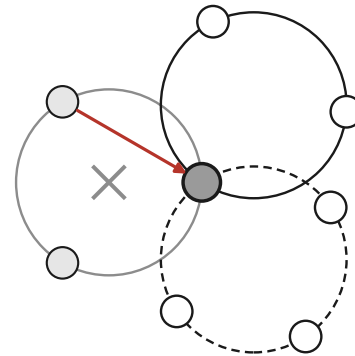
clock extended

the group transmits while any member is infectious

c

group-level branching

$$N_{ab} = C(m_b, \lambda_b, \theta_b) \left[\frac{\langle k^{(a)} k^{(b)} \rangle}{\langle k^{(a)} \rangle} - \delta_{ab} \right]$$



arriving group,
not counted again ($-\delta_{ab}$)